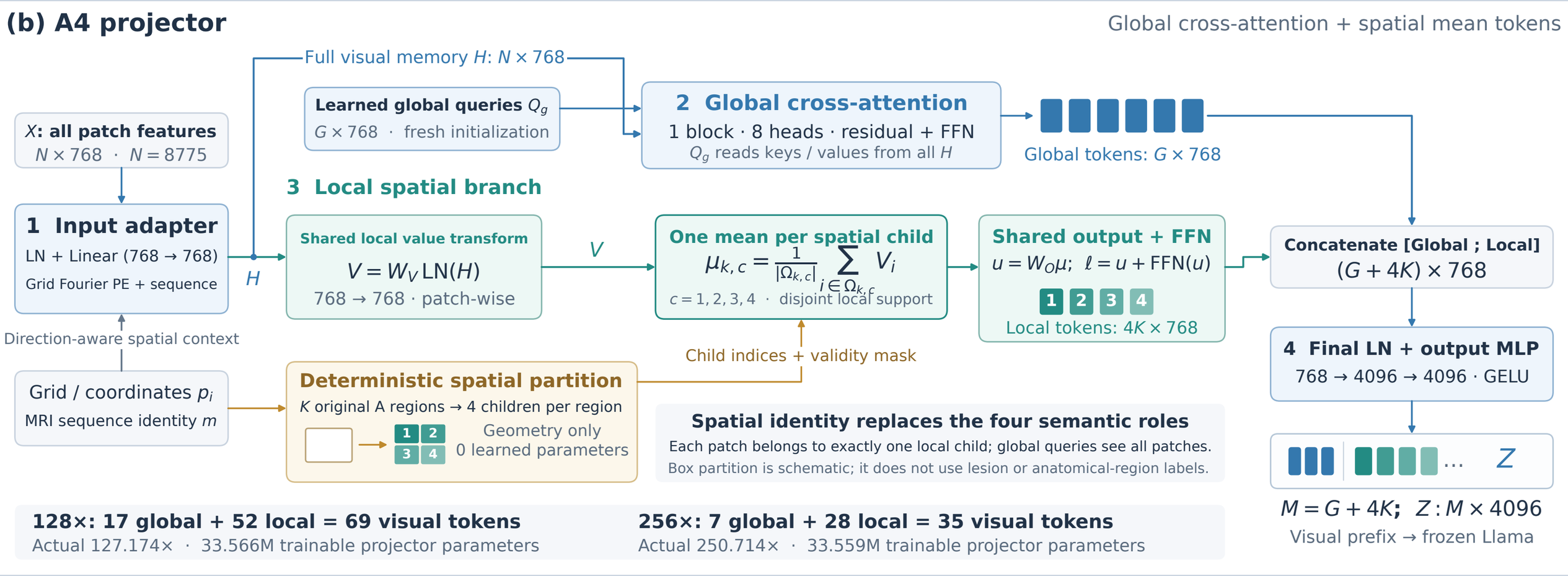
(b) A4 projector
Global cross-attention + spatial mean tokens
Full visual memory H: N × 768
X: all patch features
N × 768 · N = 8775
Learned global queries Q_g
G × 768 · fresh initialization
2 Global cross-attention
1 block · 8 heads · residual + FFN
Q_g reads keys / values from all H
Global tokens: G × 768
3 Local spatial branch
1 Input adapter
LN + Linear (768 → 768)
Grid Fourier PE + sequence
H
Shared local value transform
V = W_V LN(H)
768 → 768 · patch-wise
V
One mean per spatial child
μ_{k,c} = 1/|Ω_{k,c}| Σ_{i∈Ω_{k,c}} V_i
c = 1, 2, 3, 4 · disjoint local support
Shared output + FFN
u = W_O μ; ℓ = u + FFN(u)
1 2 3 4
Local tokens: 4K × 768
Concatenate [Global ; Local]
(G + 4K) × 768
4 Final LN + output MLP
768 → 4096 → 4096 · GELU
Direction-aware spatial context
Grid / coordinates p_i
MRI sequence identity m
Deterministic spatial partition
K original A regions → 4 children per region
1 2 3 4
Geometry only
0 learned parameters
Child indices + validity mask
Spatial identity replaces the four semantic roles
Each patch belongs to exactly one local child; global queries see all patches.
Box partition is schematic; it does not use lesion or anatomical-region labels.
Z
M = G + 4K; Z : M × 4096
Visual prefix → frozen Llama
128×: 17 global + 52 local = 69 visual tokens
Actual 127.174× · 33.566M trainable projector parameters
256×: 7 global + 28 local = 35 visual tokens
Actual 250.714× · 33.559M trainable projector parameters